DHANUSH N

AI SYSTEMS BUILDER | SOFTWARE DEVELOPER | ML & AGENTIC AI

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate and Software Developer Intern at Adya AI, building AI-enabled applications that combine LLM reasoning, retrieval-augmented generation, analytics, automation, and full-stack engineering. Experience spans client defect resolution and feature delivery, FastAPI services, conversational data assistants, streaming intelligence pipelines, computer vision, and interactive 3D visualization. Interested in turning complex data and workflows into practical, usable software.

EDUCATION

B.Tech, Artificial Intelligence and Data Science 2023 – Present
Karpagam College of Engineering, Coimbatore | CGPA: 8.72/10

Higher Secondary Certificate (HSC) 2022 – 2023
Sree Vidhya Vikas Matric Hr. Sec. School, Salem | Percentage: 94%

PROFESSIONAL EXPERIENCE

Software Developer Intern — Adya AI May 2026 – Present | Remote

Castlewise — Client Project

Role-based, multilevel platform for housing society and apartment redevelopment (live in production)

- **Tender & Business Flow:** Implemented the end-to-end tender flow and core business workflows, covering the tender lifecycle for society and apartment redevelopment projects.
- **Multilevel Role-Based System:** Built role-based access controls for screens, permissions, and actions so each stakeholder sees functionality relevant to their role.
- **Platform Modules:** Implemented the Notification module, Payment Master, and super-admin screens for user alerts, centralized administration, and payment configuration.
- **Feature Delivery:** Delivered client-requested features based on feedback while keeping changes aligned with agreed requirements.
- **Defect Resolution & QA:** Resolved Jira-tracked defects, verified fixes through testing, maintained ticket updates, and covered QA responsibilities during tester absence.
- **Client Collaboration:** Joined client meetings to clarify requirements and fix/feature questions, and shared ticket-status updates to support coordination between client and team.

RAS — Product Development

- **UI Modernization:** Redesigned and refreshed the user interface to improve usability and visual consistency.
- **Forecasting:** Fine-tuned the Data Analyst agent's forecasting capability to improve forecasts generated from user datasets.
- **3D Chart Generation:** Extended the Data Analyst agent with 3D chart generation so results can be presented as 3D visualizations instead of standard 2D charts.
- **3D Dashboard:** Built a unified 3D dashboard combining globe, bar, and line charts in one visual workspace.
- **AI Story Mode:** Developed a story mode that converts analysis results into animated slide presentations with AI voice narration.
- **Platform Modules:** Implemented Notification, Payment, and Sharing master modules for alerts, payment handling, and content sharing.

PERSONAL PROJECTS

TalentAI — Resume Intelligence Platform

Python, FastAPI, LLMs, RAG, NLP, ChromaDB, MySQL, JavaScript

- **Resume Intelligence & Matching:** Built an AI-assisted recruitment workflow that evaluates resumes against job descriptions, extracts candidate information, and ranks profiles by semantic relevance.
- **Retrieval-Augmented Ranking:** Used FastAPI, RAG, and ChromaDB to retrieve relevant candidate information and support semantic matching between candidate profiles and role requirements.
- **Profile Verification:** Integrated GitHub and LinkedIn profile checks into the candidate review workflow, helping recruiters assess resume claims alongside public professional profiles.

Flexie 2.0 — Voice-Controlled Agentic Desktop Assistant
Python, FastAPI, tool orchestration, memory, desktop automation

- **Voice & Desktop Automation:** Developing a Windows-native assistant that combines voice interaction with desktop and web actions to support hands-free task execution.
- **Agentic Task Planning:** Designed modular components for goal planning, tool registration, DAG-based task planning, workflow execution, and automation verification.
- **Memory & Tool Execution:** Structured the assistant around memory management and tool orchestration to maintain useful context and execute multi-step user workflows.

TNEA Pro AI — Engineering Admissions Counselling Assistant
FastAPI, Groq/Llama, Ollama, ChromaDB, SQLite, LangChain, Docker

- **RAG-Based Counselling:** Developed an assistant that answers questions from TNEA brochures and policy documents by combining retrieved context with structured cutoff data.
- **Cutoff Analysis & Recommendations:** Implemented cutoff calculations and Safe, Moderate, and Dream college recommendations using historical cutoff data from 2021–2025.
- **Search & Deployment:** Added searchable college profiles, choice-list exports, a TFC centre locator, pagination, short-lived query caching, request debouncing/aborting, and Docker Compose support.

AI-Based 3D Molecule Visualizer
Python, Blender Python API, HTML/CSS, 3D model export

- **Molecular 3D Pipeline:** Created a pipeline that converts molecular XYZ coordinate data into 3D structures and automates scene and model generation through Blender scripting.
- **Reusable 3D Assets:** Exported generated models as GLB assets and added a lightweight web interface for easier access to visualized molecular structures.
- **Interactive Learning:** Added voice-assisted interaction to make chemical structures easier to explore and support experiential science learning.

E-Waste Deconstructor — Computer Vision & Digital Twin
Python, YOLO, OpenCV, simulation, HTML/CSS

- **Computer Vision Classification:** Built a YOLO-based workflow to identify and classify e-waste items and estimate hazard severity to inform safer handling.
- **Digital Twin Simulation:** Created a digital-twin simulation to monitor the deconstruction workflow and provide a live view of process stages.
- **Safety Recommendations:** Connected classification results to automated safe-deconstruction recommendations surfaced through a web dashboard.

TECHNICAL SKILLS

Languages	Python, Java, Go, SQL, JavaScript
AI / ML	LLMs, Agentic AI, RAG, NLP, semantic search, LightGBM, XGBoost, Scikit-learn, TensorFlow, Keras, YOLO
Backend & APIs	FastAPI, Flask, Spring Boot, REST APIs, Streamlit
Data & Streaming	Pandas, NumPy, Apache Kafka, Pathway, data analysis, anomaly detection, forecasting
Databases & Retrieval	MySQL, SQLite, PostgreSQL, ChromaDB, Pinecone, FAISS, Snowflake
Visualization & 3D	Chart.js, Tableau, Power BI, OpenCV, Blender, Unity3D, ARCore/ARKit
Tools & Delivery	Git, Docker, Docker Compose, Jira, Jupyter Notebook, VS Code, MLOps

ACHIEVEMENTS

- **Winner** — National-Level Hackathon, IIT Kharagpur & Shiv Nadar University — Jan 2026
- **Winner** — AIM'25 Industrial Hackathon, Infynd, Coimbatore — Oct 2025
- **Winner** — HackZilla 25, KPR Institute of Technology / IEEE ComSoc — Sep 2025
- **Top Finalist** — HackBattle'25, VIT Vellore — Sep 2025
- **Top 10** — Hacknation 2026, 4th Global AI Hackathon — Feb 2026
- **Rank 33** — FIFS Sports Data, FIFO Gameathon 2.0 — Mar 2025

CERTIFICATIONS

- Education for Sustainable Development — NPTEL, Elite + Silver (86%)
- Introduction to IoT — NPTEL, Elite + Silver (76%)
- Python for Data Science — NPTEL, Elite (74%)
- Enhancing Soft Skills — NPTEL, Elite (69%)